

Rui Jin

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EDUCATION

Nanyang Technological University

Jan. 2025 – Present

Ph.D. in Electrical and Electronic Engineering

Singapore

- School of Electrical and Electronic Engineering.
- Supervised by Prof. [Lihua Xie](#).

Zhejiang University

Sep. 2021 – Mar. 2024

M.Eng. in Control Science and Engineering

Hangzhou, China

- FAST Lab, College of Control Science and Engineering.
- Co-supervised by Prof. [Fei Gao](#) and Prof. Haojian Lu.
- Postgraduate Recommendation, ranking 1/81.

Northwestern Polytechnical University

Sep. 2017 – Jun. 2021

B.Eng. in Mechanical Design & Manufacturing and Their Automation

Xi'an, China

- School of Mechanical Engineering.
- Comprehensive ranking: 1/94, top 1.1%.

INTERNSHIP EXPERIENCE

Peking University, PKU-Agibot Lab

Jun. 2024 – Dec. 2024

Research Assistant

Beijing, China

- Conducted research on affordance-based manipulation under the supervision of Prof. [Hao Dong](#).

Skysys Intelligent Technology, Research and Development Department

Aug. 2023 – Jan. 2024

Project Leader, Photovoltaic Cleaning Robot

China

- Proposed a UAV-based autonomous system for deploying and retrieving cleaning robots on photovoltaic panels.

SELECTED PUBLICATIONS

Tethered UAV Autonomous Knotting on Environmental Structures for Transport

Cyborg and Bionic Systems, 2025, IF: 18.1 | [paper](#) | [pdf](#)

- **Rui Jin***, Xinhang Xu*, Yizhuo Yang, Jianping Li, Muqing Cao, Lihua Xie.
- Proposed a tethered UAV-based autonomous knotting system for cable anchoring and heavy-load transport in unstructured environments.
- Developed Knot Planner, integrating human-guided enclosing plane extraction, frontier-based enclosing path search, and knot-specific trajectory optimization.
- Validated the system through real-world outdoor experiments and simulation studies, demonstrating autonomous knotting on unseen structures and lifting a 15.3 kg payload.

Unmanned Aerial Vehicle Mediated Drug Delivery for First Aid

Advanced Materials, 2023, Frontispiece, IF: 29.4 | [pdf](#) | [paper](#)

- Sheng Tao*, **Rui Jin*** (co-first author), et al., Fei Gao, Haojian Lu, Jichen Yu, Zhen Gu.
- Proposed a UAV-mediated first-aid system enabling autonomous administration of emergency medication without bystander or conscious-patient involvement.
- Designed a contact-triggered microneedle applicator that provides sufficient insertion force upon contact with skin.

GS-Planner: A Gaussian-Splatting-based Planning Framework for Active High-Fidelity Reconstruction

IROS 2024 | [pdf](#) | [video](#) | [arxiv](#)

- **Rui Jin***, Yuman Gao*, Haojian Lu, Fei Gao.
- Proposed the first active 3D reconstruction system using 3D Gaussian Splatting with online evaluation.
- Designed an online model-consistent completeness and quality evaluation feedback strategy.
- Devised a planning framework for active reconstruction and safe navigation in the 3DGS map.

QuadHand: A CoG-Compensated Quadrotor Aerial Manipulator with Whole-Body Motion Planning

Submitted to *IROS 2026*

- **Rui Jin***, Ruiyang Liu*, Xinhang Xu, Yi Wang, Haotian Jin, Yizhuo Yang, Lihua Xie.
- Designed QuadHand, a compact quadrotor aerial manipulator with a 3-DOF arm, gripper, and passive center-of-gravity compensation mechanism.
- Proposed a multi-articulated robot-centric signed distance field representation to preserve geometric fidelity for close-proximity aerial manipulation.
- Developed a spatiotemporal whole-body trajectory optimization framework for coupled base-arm planning in cluttered environments.

DuctAM: A Duct-Assisted Quadrotor-Based Aerial Manipulator Enabling High-Force Push-and-Pull Interactions

Submitted to *IROS 2026*

- Yi Wang*, **Rui Jin*** (co-first author), Xinhang Xu*, Haotian Jin, Ruiyang Liu, Yizhuo Yang, Lihua Xie.
- Proposed DuctAM, a duct-assisted quadrotor aerial manipulator for high-force push-and-pull interaction tasks.
- Developed an acceleration decomposition-based control scheme to decouple longitudinal force generation from pitch motion during sustained contact.
- Validated the platform in real-world interaction tasks, including cart pushing, door closing, drawer opening, and high-friction crate pushing.

OTHER PUBLICATIONS

- **HGS-Planner: Hierarchical Planning Framework for Active Scene Reconstruction using 3D Gaussian Splatting.** Zijun Xu, **Rui Jin**, et al., Zhongxue Gan, Wenchao Ding. *ICRA*, 2025.
- **Canfly: A Can-sized Autonomous Mini Coaxial Helicopter.** Neng Pan, **Rui Jin**, Chao Xu, Fei Gao. *IROS*, 2023.
- **Hand-like autonomous flying robot for airborne grasping and interaction.** Yuze Wu*, Fan Yang*, **Rui Jin**, Yuhang Zhong, Junjie Wang, Fei Gao. *Nature Communications*, 2026.
- **Adaptive Tracking and Perching for Quadrotor in Dynamic Scenarios.** Yuman Gao, Jialin Ji, Qianhao Wang, **Rui Jin**, Yi Lin, Zhimeng Shang, Shaojie Shen, Chao Xu, Fei Gao. *T-RO*, 2024.
- **Modeling and Force Control of a Variable-Length Continuum Robot with Variable Stiffness for Minimally Invasive Surgery.** Jingyu Zhang, Qin Fang, Lili Liu, **Rui Jin**, Pingyu Xiang, Rong Xiong, Yue Wang, Haojian Lu. *T-ASE*, 2024.
- **Soft Lightweight Small-Scale Parallel Robot With High-Precision Positioning.** Qin Fang, Jingyu Zhang, et al., **Rui Jin**, Yue Wang, Rong Xiong, Zhefeng Gong, Haojian Lu. *T-MECH*, 2023.
- **A Survey on Design, Actuation, Modeling, and Control of Continuum Robot.** Jinyu Zhang, Qin Fang, et al., **Rui Jin**, Ke Qiu, Yue Wang, Rong Xiong, Haojian Lu. *Cyborg and Bionic Systems*, 2022.
- **Fast Iterative Region Inflation for Computing Large 2-D/3-D Convex Regions of Obstacle-Free Space.** Qianhao Wang, Zhepei Wang, Mingyang Wang, et al., **Rui Jin**, Yuman Gao, Chao Xu, Fei Gao. *T-RO*, under review.

RESEARCH EXPERIENCES

Design, Modeling and Control of Miniature Coaxial Helicopter

Dec. 2020 – Nov. 2023

Master's Thesis

- Proposed the hardware design and flight control algorithm for an autonomous coaxial dual-rotor UAV weighing 1.5 kg with a 33-minute endurance.
- Developed a differential-flatness-based cascading controller and a quadratic-programming-based actuator allocation algorithm to prevent actuator saturation.

Neural Collision Field for Efficient Trajectory Optimization for Mobile Robots

Jan. 2023 – Present

Research Project

- Compressed a swept-volume-based continuous-time signed distance field into a neural network for efficient and accurate representation.
- Applied the learned representation to quadrotor trajectory optimization.
- Designed a progressive guidance training strategy and a level-of-details sampling strategy to enhance network performance and training efficiency.

Nuclear Power Plant Autonomous Inspection Tracked Robot

Aug. 2023 – Dec. 2023

Research Project

- Engineered software algorithms for autonomous inspection, obstacle avoidance, and cross-platform operation of a nuclear power plant inspection tracked vehicle.

COMPETITION EXPERIENCES

China Robot Competition

2019

Championship Prize, Top 1%

- Designed and built a transformer robot capable of transforming between vehicular and humanoid forms.

National 3D Innovative Design Competition

2019

Top-tier Award of Shanxi Province, Top 3%

HONORS AND AWARDS

- Outstanding Graduate of Northwestern Polytechnical University, 2021.
- Interdisciplinary Contest In Modeling, Honorable Mention, 2020.
- Fastgear Scholarship, Top-tier Prize, Top 1%, 2019.
- National Training Program of Innovation for Undergraduates, Outstanding Conclusion Award, 2018 and 2019.
- Huawei Scholarship, First Prize, Top 1%, 2018.
- Northwestern Polytechnical University Scholarship, First Prize, Top 10%, 2017, 2018, and 2019.

SKILLS

Programming Skills: C++, Python, MATLAB, ROS

English Proficiency: IELTS 7.0, GRE 327, CET-6 581

Design Skills: SolidWorks, Altium Designer, KeyShot, Premiere

Video Production: FAST Lab video account management; produced video edits with over one million views